D03 · Production Quality & Compliance Systems

03 · CAPA — Corrective & Preventive Action

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

CAPA is the structured loop between a quality problem (or potential problem) and the process change that prevents recurrence — the corrective for what already happened, the preventive for what could. Mature CAPA systems are triggered by internal nonconformances, customer complaints, audit findings, supplier issues, and SPC violations. The methodology is standardized (8D, A3, ISO 13485 §8.5, 21 CFR 820.100); the discipline is whether root cause is real and whether action is verified.

2 WHY IT MATTERS

CAPA is the only quality mechanism that converts symptom into structural change — and the only one that survives auditor scrutiny while being almost universally gamed. **The dominant failure mode is the CAPA closed without verifiable root cause, which guarantees the same problem will return under a different file number within months**. The foundational text on Japanese quality methodology documents the discipline that separates real root cause from causal speculation **(Ishikawa, 1985 — peer-reviewed, "What is Total Quality Control? The Japanese Way")**: the 5-why is not a checklist but a structured chain of inference, each step verified independently by data, experiment, or controlled replication. Industry data on CAPA effectiveness is consistent — **operations where 60%+ of closed CAPAs have verified root cause see issue recurrence below 10%; operations below 30% verified root cause see recurrence above 40%**. The asymmetry is brutal: a poorly executed CAPA system produces the same defects with new paperwork on each occurrence, consuming engineering hours while preventing nothing.

3 HOW IT IS DONE

1. **Trigger CAPA at the right threshold — not every nonconformance is a CAPA candidate.** The most common pathology is making every NCR a CAPA, which buries the system in low-value records and starves attention from the high-impact ones. Define explicit triggers: customer complaints, repeat NCRs of the same failure mode, FMEA RPN above threshold, audit findings of major+ severity, SPC special-cause patterns three times in 30 days. CAPA is for systemic problems; the rest belong in disposition logs.
2. **Lead with containment before investigation — the customer does not wait for your 5-why.** D3 of the 8D framework exists for a reason: when a defect ships, the first move is to stop additional shipments, segregate suspect stock at customer and in-house, and replace product. Investigation happens in parallel, not first. The most common audit finding on CAPA execution is investigation that started before containment, leaving the customer exposed while engineering debated root cause.
3. **Treat root cause as a falsifiable claim, not a worksheet entry.** A 5-why with five answers but no evidence is causal speculation, not analysis. Every "why" must be verified — by production data, by designed experiment, by replication of the failure mode in controlled conditions. A root cause is the cause whose removal eliminates the failure; if you cannot prove removal eliminates it, you have not found root cause yet, regardless of how confident the team feels.
4. **Implement corrective action that is detectable in the absence of the failure.** The corrective action is real when you can audit its existence months later without depending on the failure to confirm it — procedure updated and operators trained, FMEA updated, control plan modified, poka-yoke implemented, supplier specification changed. Actions phrased as "we will be more careful" or "operators were reminded" are not actions; they are aspirations dressed as closures.
5. **Verify effectiveness with a defined sample size and timeframe — not with "we will keep an eye on it."** Effectiveness verification requires production volume sufficient to detect recurrence with statistical confidence, typically 30+ days or N+ units depending on the baseline failure rate. CAPAs closed before verification is complete are CAPAs that get reopened — and reopened CAPAs are the leading indicator of a CAPA system that has lost its discipline.
6. **Trend the CAPA portfolio quarterly — repeat causes are signals about the underlying system.** When the same root cause appears across multiple CAPAs (training gap, supplier control gap, equipment maintenance gap, design rule gap), the system has a structural issue that no individual CAPA will resolve. The quarterly CAPA trend review is where the operating system signals its own diagnosis — and where the operations leadership decides whether to address a class of problems or

continue managing them one at a time.

WORKED EXAMPLE — ILLUSTRATIVE

A medical-device manufacturer operating under 21 CFR 820 across 4 product families has a CAPA system carrying **450 open records, 80% lacking documented root-cause verification, 31% twelve-month recurrence rate, and 47% on-time closure**. The 2023 FDA inspection produces a Form 483 with 14 observations, most CAPA-related. The implementation: a 9-month transformation establishes strict 8D structure with mandatory peer review at the root-cause checkpoint; CAPA triggers tightened to remove low-value records (NCRs without systemic implication move to disposition logs); dedicated effectiveness-verification protocol with predefined sample sizes per failure class; quarterly trend analysis treated as a formal management review input. Results at 12 months: **open CAPAs from 450 to 78, recurrence from 31% to 4.7%, on-time closure from 47% to 91%, FDA observations on next inspection down to 2**. The number of NEW CAPAs opened per quarter actually rises 22% — the system now generates honest signal instead of consuming it. The leadership lesson is counter-intuitive: more CAPAs opened, fewer recurring problems.

IN PRACTICE

Treat root cause as a falsifiable claim

The 5-why is not a checklist but a structured chain of inference — each step verified independently by data, experiment, or controlled replication **(Ishikawa, 1985 — peer-reviewed)**. The "human error" root cause that does not survive falsification is not a root cause.

Explicit co-ownership: methodology, investigation, action, verification

QE owns methodology and trend analysis; operations owns the investigation and action; engineering owns design changes; the cell supervisor owns effectiveness verification. The CAPA without a single accountable owner is the CAPA that drifts open for 18 months and closes the day before the audit.

Interrogate the system, not the operator

Most root causes that appear to be "human error" are latent organizational conditions — training adequacy, ergonomics, supervisory pressure, system design **(Reason, 1990 — Cambridge University Press)**. An investigation focused on the operator will miss these by design.

Verify effectiveness with defined sample sizes; trend the portfolio quarterly

Plants with >60% verified-root-cause CAPAs see recurrence below 10%; plants below 30% see recurrence above 40%. Without quarterly trending of the CAPA portfolio, systemic patterns hide in case-by-case file numbers.

WHAT YOU WOULD USE

CAPA PLATFORMS

MasterControl CAPA / Veeva Vault QualityDocs — Enterprise CAPA modules for FDA-regulated industries (pharma, medical device, biotech) with deep 21 CFR 820 / ISO 13485 compliance, predicate-rule audit trails, and validated electronic signature workflows **(MasterControl — vendor; Veeva — vendor)**.

Sparta TrackWise / ETQ Reliance — Broad enterprise CAPA platforms across pharma, automotive (IATF 16949), aerospace (AS9100), and general manufacturing; configurable workflow, integrated trend analytics, and supplier CAPA integration **(Sparta Systems — vendor; ETQ — vendor)**.

Greenlight Guru / Qualio / Arena (PTC) — Mid-market CAPA platforms tuned for medical-device and growing life-sciences operations — faster deployment, lower TCO than the enterprise tier, with workflow templates aligned to ISO 13485 and 21 CFR 820 out of the box **(Greenlight Guru — vendor; Qualio — vendor)**.

Structured 8D in Smartsheet / Jira / Power Apps with a CAPA register in SharePoint — For plants without QMS platform budget — works under ISO 9001 if the discipline of root-cause verification and trend analysis is maintained by a dedicated quality engineer. Fails in regulated environments and fails the moment the dedicated owner leaves.

4 SO WHAT

The CAPA closed without verified root cause is paperwork with a number on it — not corrective, not preventive, certainly not action. Trigger at the right threshold, contain before investigating, treat root cause as a falsifiable claim, implement actions detectable in the absence of failure, verify effectiveness with defined sample sizes, and trend the portfolio quarterly to surface systemic patterns. The CAPA system is the only quality mechanism that converts symptom into structural change; ungoverned, it converts symptom into recurrence with new paperwork.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México